

P5.2 The electromagnetic spectrum

Summary

Having looked at mechanical waves, waves in the electromagnetic spectrum are now considered. This section includes the application of electromagnetic waves with a specific focus on the behaviour of light. Alongside this, it explores the application of other types of electromagnetic radiation for use in medical imaging.

Underlying knowledge and understanding

Learners may be familiar with uses of some types of radiation but an understanding of all parts of the electromagnetic spectrum is not expected and should be taught as new content.

Common misconceptions

Learners can have misconceptions such as gamma rays, X-rays, ultraviolet light, visible light, infrared light, microwaves and radio waves being independent entities and not being able to view it as a spectrum. They struggle to link the features that waves have in common, alongside the differences and how these relate to their different properties.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
P5.2a	recall that electromagnetic waves are transverse and are transmitted through space where all have the same velocity				
P5.2b	explain that electromagnetic waves transfer energy from source to absorber	examples from a range of electromagnetic waves			
P5.2c	apply the relationships between frequency and wavelength across the electromagnetic spectrum		M1a, M1c, M3c	WS1.1b, WS1.3b, WS1.3e	Investigation of electromagnetic waves on chocolate or processed cheese in a microwave to measure wavelength. (PAG P4)
P5.2d	describe the main groupings of the electromagnetic spectrum and that these groupings range from long to short wavelengths and from low to high frequencies	radio, microwave, infrared, visible (red to violet), ultraviolet, X-rays and gamma rays		WS1.1c, WS1.1d, WS1.1e, WS1.1f, WS1.1h, WS1.1i	Research and design a poster to show the properties, uses and dangers of the different electromagnetic wave groups.
P5.2e	describe that our eyes can only detect a limited range of the electromagnetic spectrum				

P5.2f	recall that light is an electromagnetic wave				
P5.2g	give examples of some practical uses of electromagnetic waves in the radio, microwave, infrared, visible, ultraviolet, X-ray and gamma ray regions			WS1.1b, WS1.1d, WS1.1e, WS1.1f, WS1.1h, WS1.1i, WS1.3e, WS1.3f	Demonstration of how microwaves can be used to light a bulb in a beaker of water and of how this shows that microwaves heat water in foods. Use a microwave emitter and absorber to demonstrate behaviour of waves. (PAG P8) Use of a phone camera to look at the infrared emitter on a remote control. (PAG P8)
P5.2h	describe how ultraviolet waves, X-rays and gamma rays can have hazardous effects, notably on human bodily tissues			WS1.1a, WS1.1c, WS1.1d, WS1.1e, WS1.1f, WS1.1h, WS1.1i	Show images of X-rays to discuss how the images are formed; their advantages and disadvantages. Investigation of the balance of risks for staff and patients during radiotherapy.
P5.2i ☒	explain, in qualitative terms, how the differences in velocity, absorption and reflection between different types of waves in solids and liquids can be used both for detection and for exploration of structures which are hidden from direct observation, notably in our bodies	the use of infrared, X-rays, gamma rays and ultrasound as an alternative in medical imaging			
P5.2j	recall that radio waves can be produced by, or can themselves induce, oscillations in electrical circuits				